Advanced Mechanics of Materials

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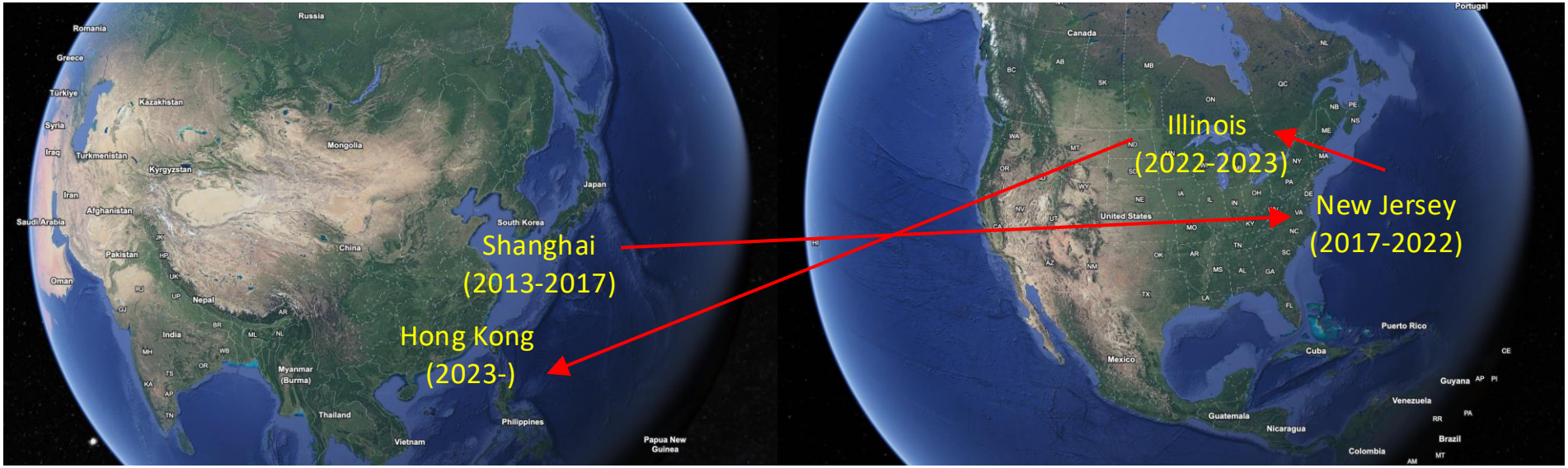
A little bit about the instructor…



Instructor:

- **Tianju XUE**

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Major author of **JAX-FEM**: an open-source finite element library.

JAX-FEM

https://github.com/deepmodeling/jax-fem

Syllabus

Date	
5 Feb 2026	Lec01: Introduction; Mathematical preliminary
12 Feb 2026	Lec02: Kinematics (deformation)
26 Feb 2026	Lec03: Kinematics (strain)
5 Mar 2026	Lec04: Kinematics (rate of strain); Stress
12 Mar 2026	Lec05: Balance principles
19 Mar 2026	Lec06: Thermal dynamics (1st Law)
26 Mar 2026	Lec07: Variational principles
2 April 2026	Lec08: Thermal dynamics (2nd Law)
9 April 2026	Lec09: Objectivity
16 April 2026	Lec10: Constitutive laws
23 April 2026	Lec11: Advanced topic I: Fracture mechanics
30 April 2026	Lec12: Advanced topic II: Stability of structures
7 May 2026	Lec13: Advanced topic III: Homogenization

Useful information

Lectures

- Room 2612B
- Thursday: 09:00 – 11:50

Grading

- Participation: 10%
- Assignments: 40%
- Final Exam: 50%

References

Lec01 - Lec10 (Standard continuum mechanics):

- Holzapfel, Gerhard A. *"Nonlinear solid mechanics: a continuum approach for engineering science."* (2002): 489-490.
- Gurtin, Morton E., Eliot Fried, and Lallit Anand. *"The mechanics and thermodynamics of continua."* Cambridge university press, 2010.

References (continued)

Lec11 (Fracture mechanics):

- Bazant, Zdenek P., and Jaime Planas. "*Fracture and size effect in concrete and other quasibrittle materials*". Routledge, 2019.

Lec12 (Stability of structures):

- Bazant, Zdenek P. and Cedolin, Luigi. "*Stability of structures: elastic, inelastic, fracture and damage theories*". World Scientific, 2010.
- https://comet-fenics.readthedocs.io/en/latest/demo/buckling_3D/buckling_3d_solid.html

Lec13 (Homogenization):

- Schröder, Jörg. *"A numerical two-scale homogenization scheme: the FE2-method."* In *Plasticity and beyond: microstructures, crystal-plasticity and phase transitions*, pp. 1-64. Vienna: Springer Vienna, 2014.
- Kalamkarov, Alexander L., and Krishna S. Challagulla. *"Effective properties of composite materials, reinforced structures and smart composites: asymptotic homogenization approach."* In *Effective properties of heterogeneous materials*, pp. 283-363. Dordrecht: Springer Netherlands, 2013.